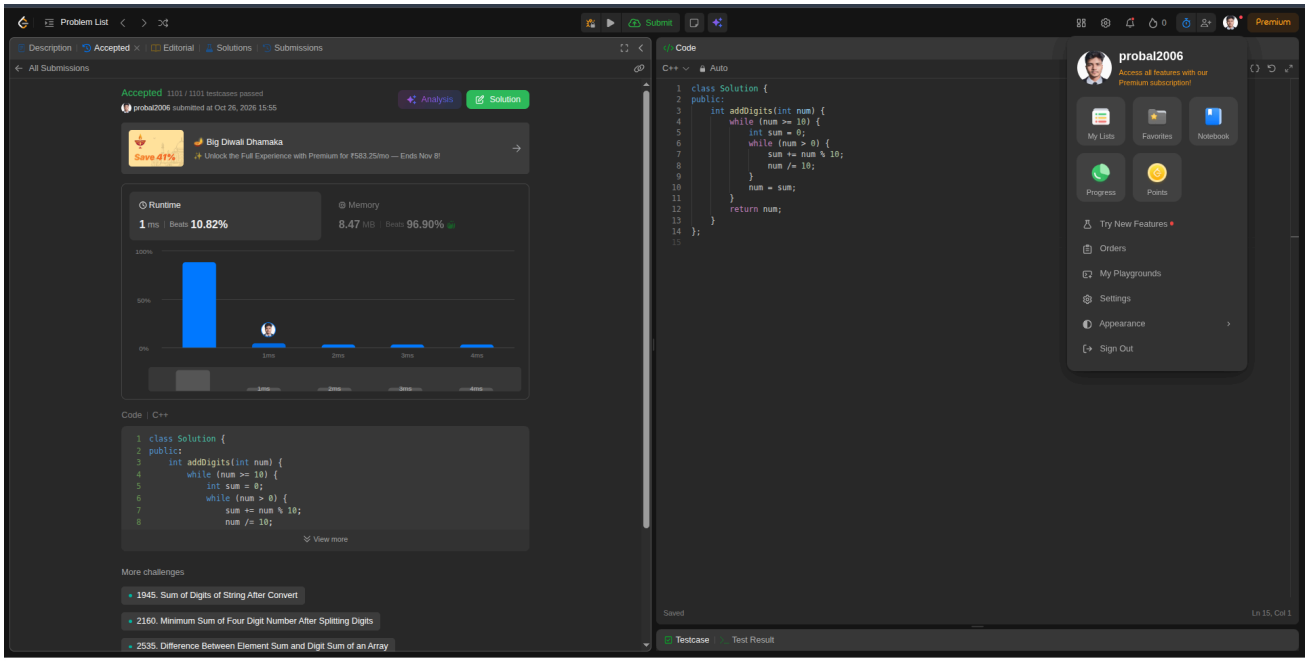
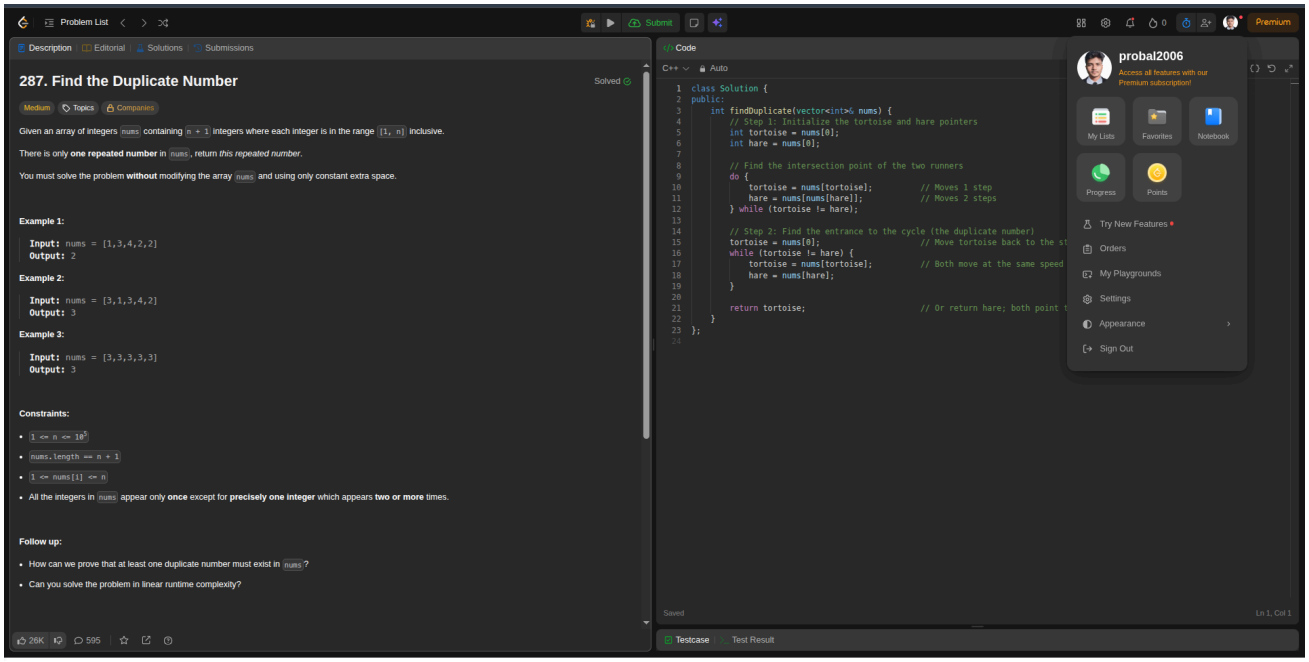

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probal2006 submitted at Oct 26, 2026 15:55
Analysis
Solution
Big Diwali Dhamaka
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Runtime
1 ms | Beats 10.82%
Memory
8.47 MB | Beats 96.90%
Code | C++
class Solution {
public:
int addDigits(int num) {
while (num >= 10) {
int sum = 0;
while (num > 0) {
sum += num % 10;
num /= 10;
}
num = sum;
}
return num;
}
};
View more
More challenges
1945. Sum of Digits of String After Convert
2160. Minimum Sum of Four Digit Number After Splitting Digits
2535. Difference Between Element Sum and Digit Sum of an Array
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C++
Auto
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Problem List
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287. Find the Duplicate Number
Solved
Medium
Topics
Companies
Given an array of integers nums containing n + 1 integers where each integer is in the range [1, n] inclusive.
There is only one repeated number in nums, return this repeated number.
You must solve the problem without modifying the array nums and using only constant extra space.
Example 1:
Input: nums = [1,3,4,2,2]
Output: 2
Example 2:
Input: nums = [3,1,3,4,2]
Output: 3
Example 3:
Input: nums = [3,3,3,3,3]
Output: 3
Constraints:
1 <= n <= 10^5
nums.length == n + 1
1 <= nums[i] <= n
All the integers in nums appear only once except for precisely one integer which appears two or more times.
Follow up:
How can we prove that at least one duplicate number must exist in nums?
Can you solve the problem in linear runtime complexity?
26K
595
Code
C++
Auto
class Solution {
public:
int findDuplicate(vector<int>& nums) {
// Step 1: Initialize the tortoise and hare pointers
int tortoise = nums[0];
int hare = nums[0];
// Find the intersection point of the two runners
do {
tortoise = nums[tortoise]; // Moves 1 step
hare = nums[nums[hare]]; // Moves 2 steps
} while (tortoise != hare);
// Step 2: Find the entrance to the cycle (the duplicate number)
tortoise = nums[0]; // Move tortoise back to the st
while (tortoise != hare) {
tortoise = nums[tortoise]; // Both move at the same speed
hare = nums[hare];
}
return tortoise; // Or return hare; both point t
}
};
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